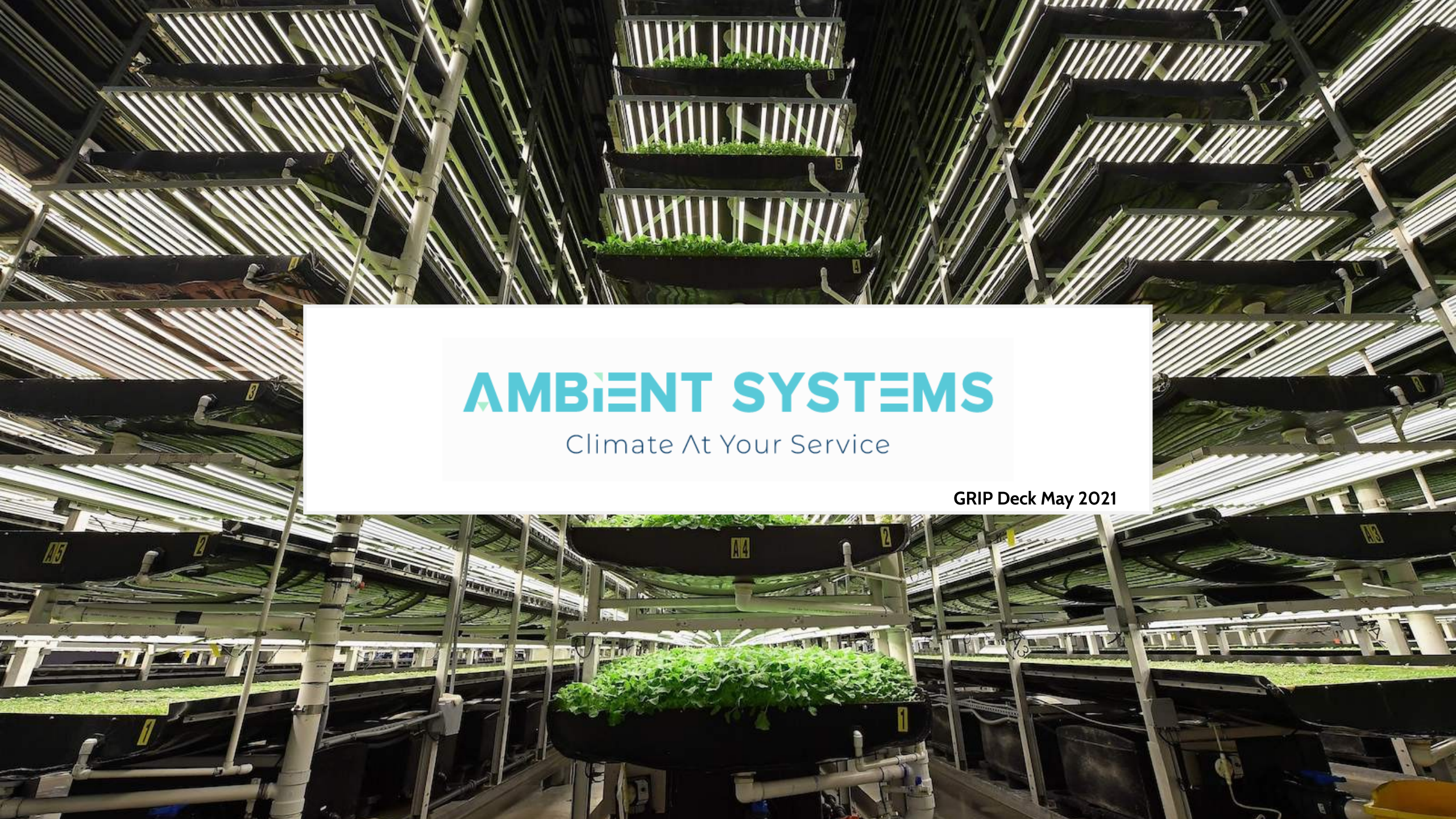


AMBIENT SYSTEMS

Climate At Your Service

GRIP Deck May 2021

Building needs have becomes increasingly complex

Indoor Farms

Modern buildings with specific applications require more than

Healthcare

Restaurants

er

Getting the most out of operating a vertical farm

Utilities
$3



Rent
$30



Human Capital
$300

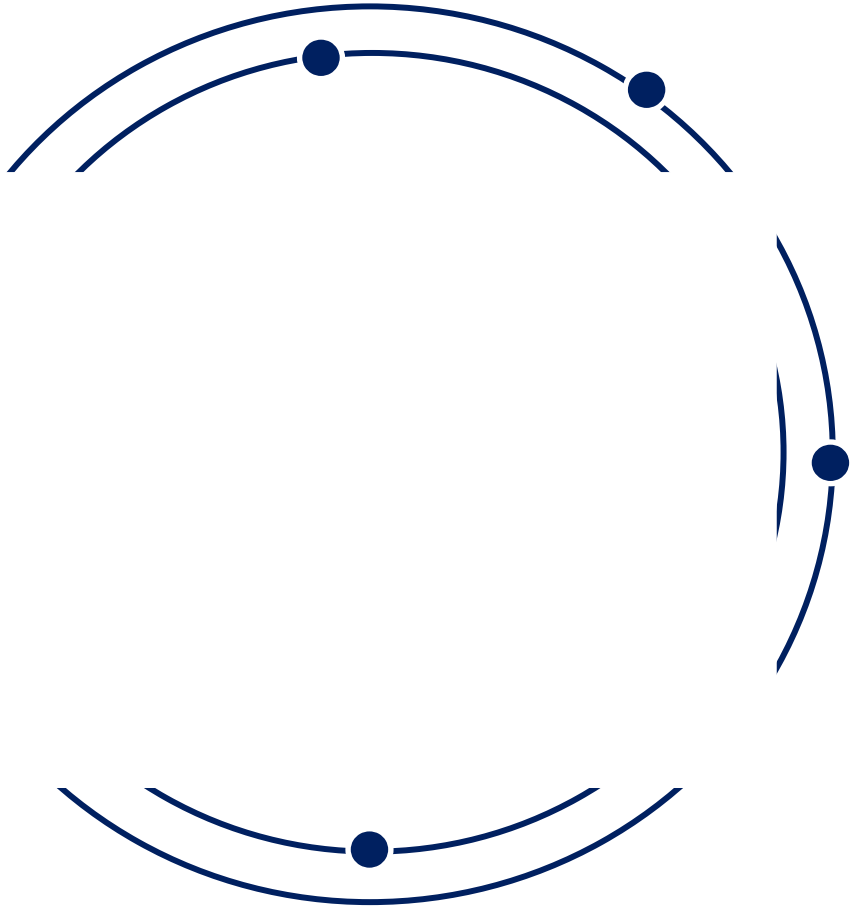


Source: Jones

Private and Confidentia

AMBIENT SYSTEMS
Climate At Your Service

Climate Management Platform for Indoor Farms



Connected.

Centralized.

Automated.


A5
2
A4
2
A3
1
1

Understanding relationship between building conditions and crops

Temperature

Soil PH level

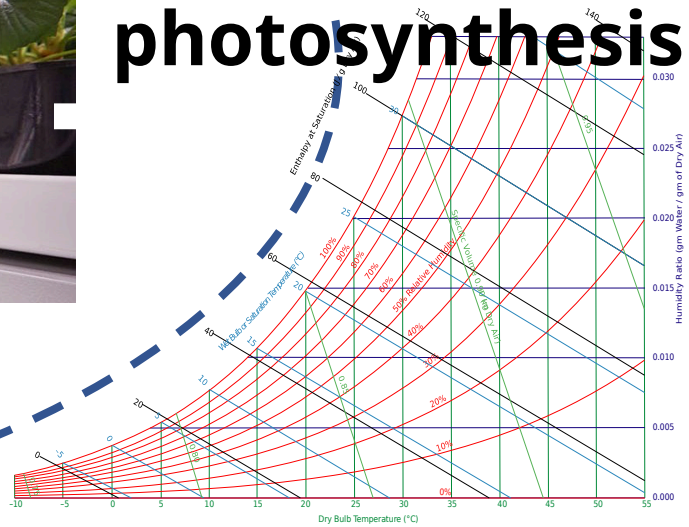
Lighting

Nutrients

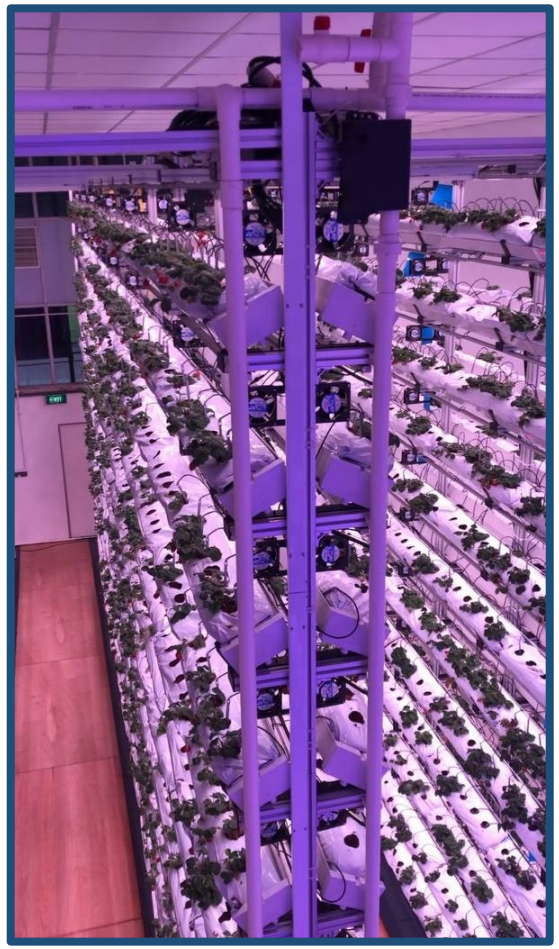


Humidity

Conditions outside of micro-ecosystem affect level of photosynthesis



Undesired interactions between buildings and crops



Strawberries growing in varying levels of

Temperature
Humidity
CO2 Levels





Habitat for pests

AMBIENT SYSTEMS
Climate At Your Service

This is what Ambient will look like within 2 years

AMBIENT SYSTEMS
Climate At Your Service

First 18 months

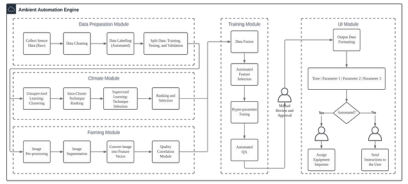
Next 6 months

Sense



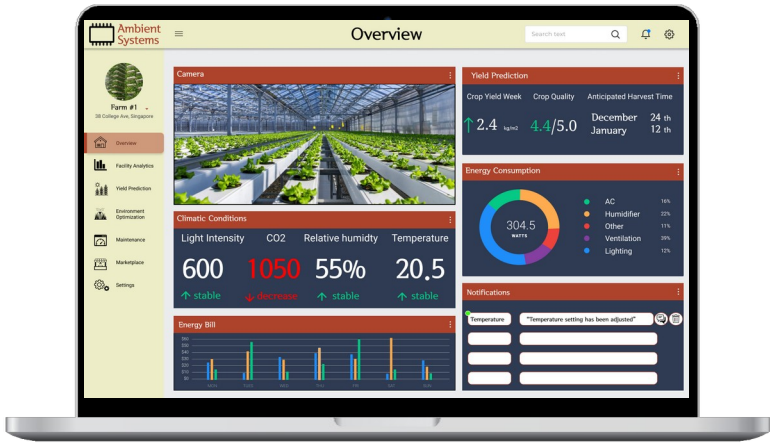
Smart Sensors

Optimize



Automation Engine

Monitor



Dashboard

Control



Edge Nodes & Control units

Patent G06N: Computer Systems Based On Specific Computational Models

Three Stage Supervised Learning

X: Equipment Data

Y: Climate Data

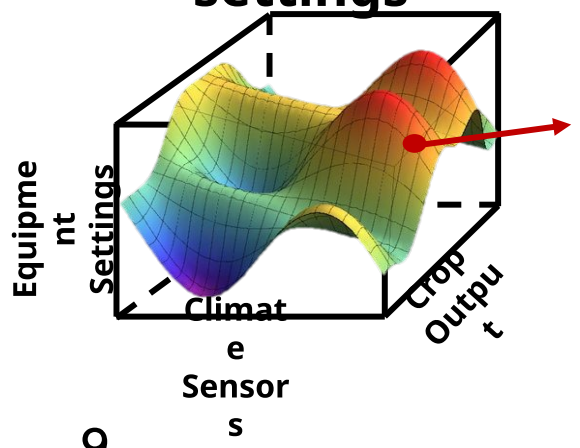
Predict Climate-based on Settings

X: Climate Data

Y: Crop Data

Predict Output based on Climate

Predict Output based on settings



Optimal Point

AMBIENT SYSTEMS
Climate At Your Service

Building conditions are automated according to generated insights

Climate Data

	Parameter
Current POC	Air Temperature
	Relative Humidity
	CO2
	Spectrum Light

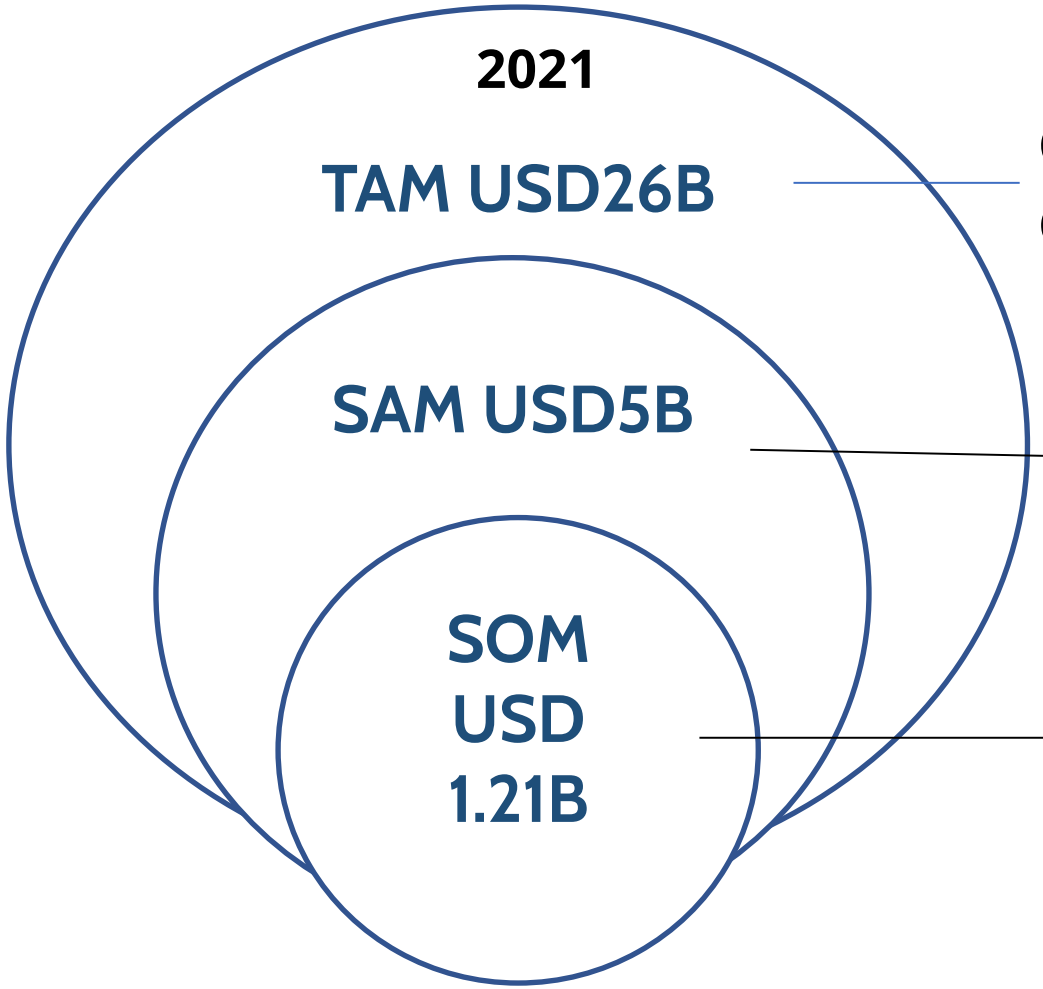
Crop Data

	Parameter
Full-scale Plot	Soil Temperature
	Soil Moisture
	pH level
	Electrical conductivity (EC_w)
	Crop specific parameters

Equipment Settings

Sample Action
• Temperature setpoint • Adjust ventilation • Install fogging machines • Install CO2 regulator • Adjust non-farming lights • Increase light intensity

Market Opportunity: TAM, SAM, SOM



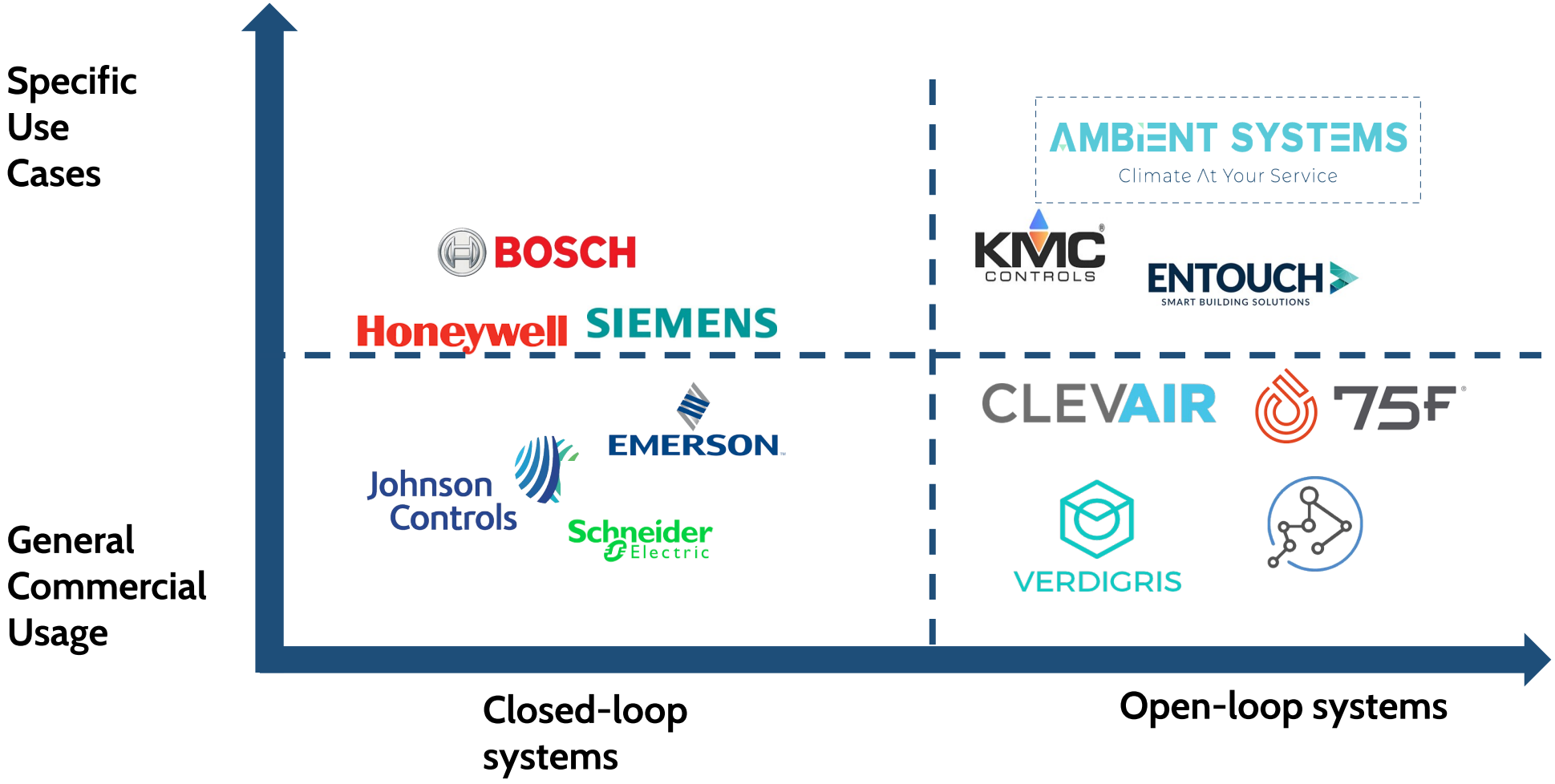
Global HVAC Automation and Controls

AI in Building Automation

APAC
Vertical Farming

Source: Frost & Sullivan, McKinsey, Data Bridge

We provide value beyond building controls and energy savings

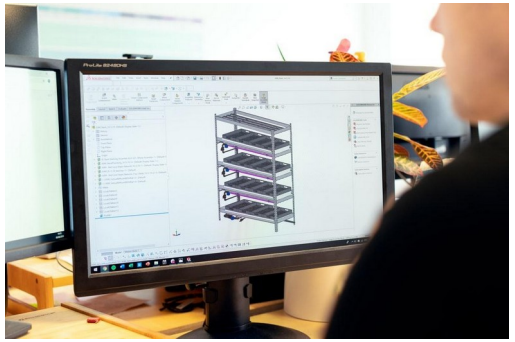


Source: Honeywell

For farmers that are looking to collect and monitor building data

Yearly Subscription Model

Basic Monitoring



One-time

Installing Sensors
SGD1.12 / Sq ft

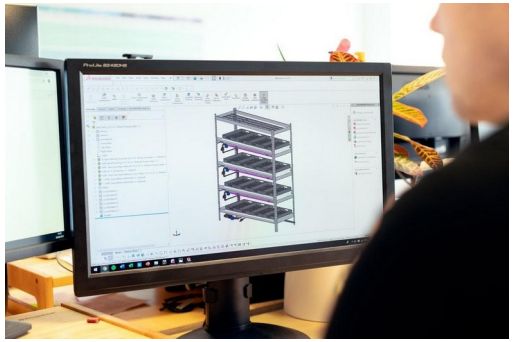
Per Month

Software Subscription
SGD500 / month

- **Real-time data**
- **Online dashboards**
- **Push Alerts**

For farmers that are looking to automate their building conditions

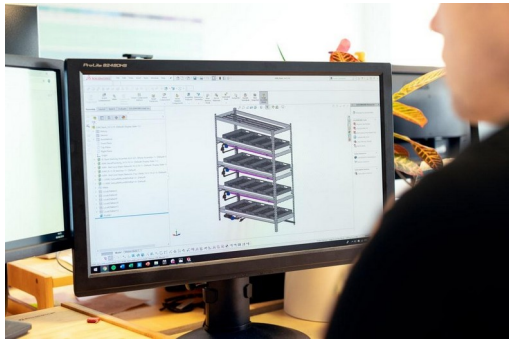
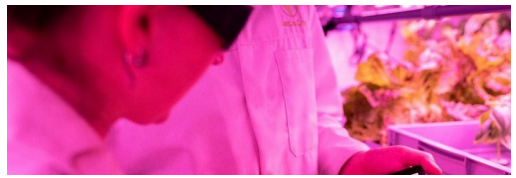
Yearly Subscription Model

	Basic Monitoring	Automating Building Controls
		
One-time	Installing Sensors SGD1.12 / Sq ft	Integration Cost SGD2.23 / Sq ft
Per Month	Software Subscription SGD500 / month	Cost of Maintenance $500 / month

- **Full HVAC connectivity**
- **Real-time environmental adjustments**

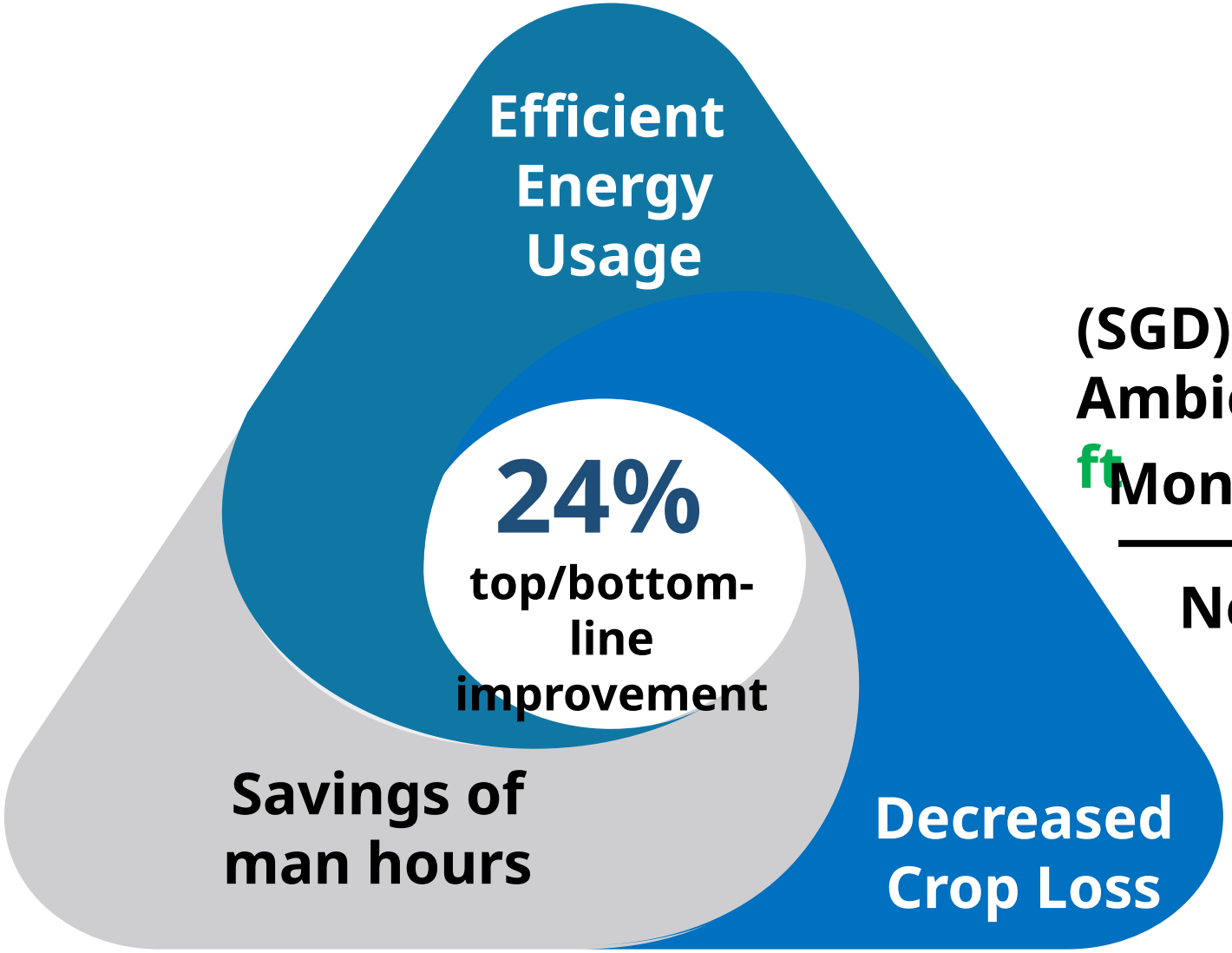
For farmers that are looking to integrate crop and building data

Yearly Subscription Model

	Basic Monitoring	Automating Building Controls	Advanced Insights
			
One-time	Installing Sensors SGD1.12 / Sq ft	Integration Cost SGD2.23 / Sq ft	• Integrate user data • Obtain Actionable Insights according to needs
Per Month	Software Subscription SGD500 / month	Cost of Maintenance SGD500 / month	Additional Insights SGD300 / month

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Farmers can expect to break-even installation costs in 4-6months



(SGD)

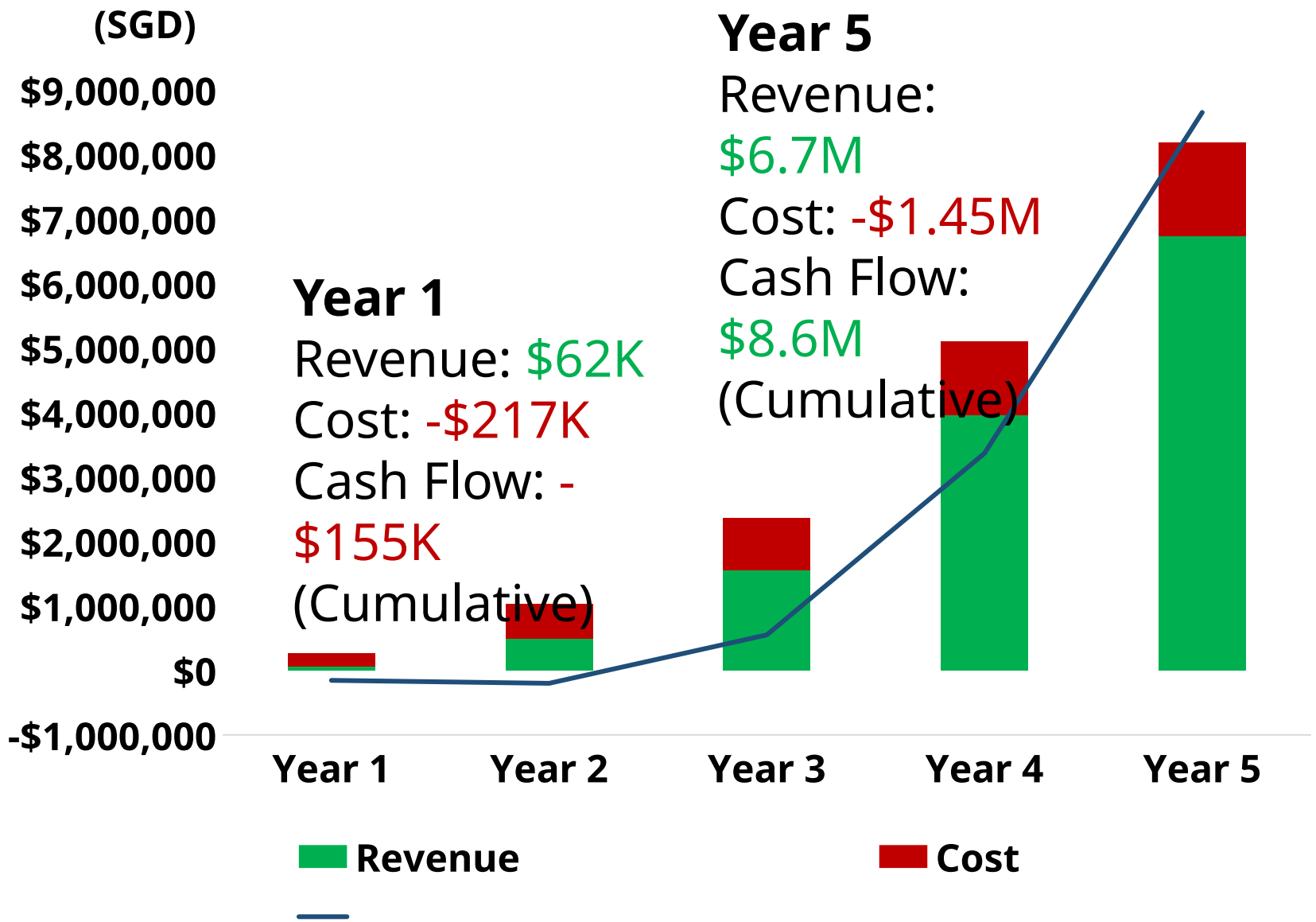
Ambient Value: $4.35 per sq ft

Monthly Cost: ($2.66 per sq ft)

Nett Value: $1.69 per sq ft

AMBIENT SYSTEMS
Climate At Your Service

Our financial projections



81.2%
Average Gross Profit

$8.5M
NPV

Private and Confidential

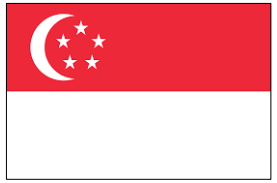
AMBIENT SYSTEMS
Climate At Your Service

We have a funding gap of SGD252K over the next 18 months

July 2021

Dec 2022

Market Expansion

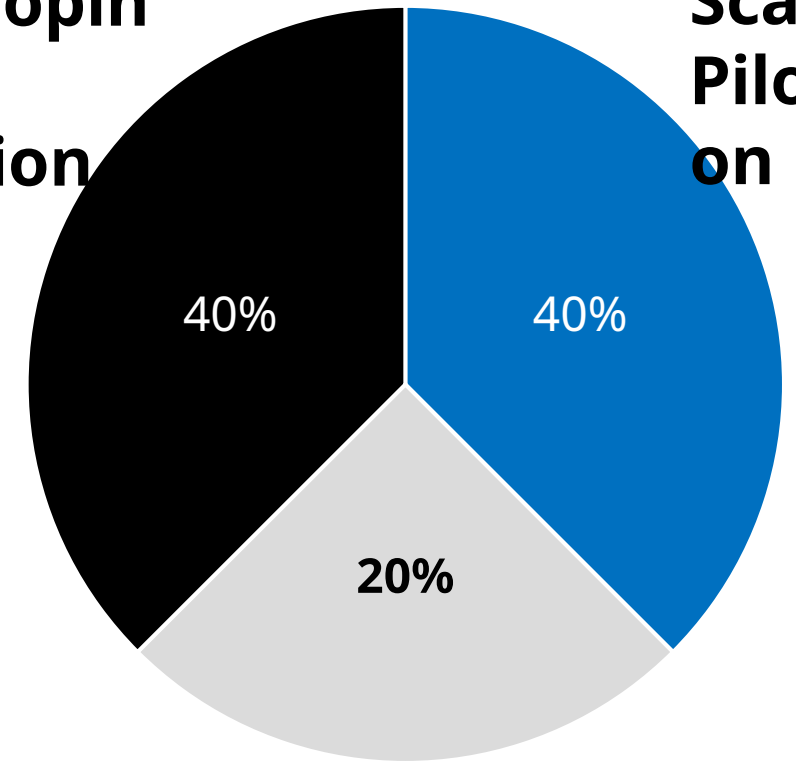


Offerings

- Monitoring
- User Insights

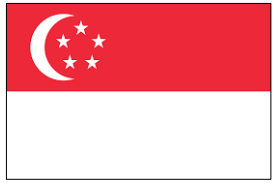
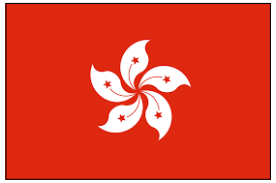
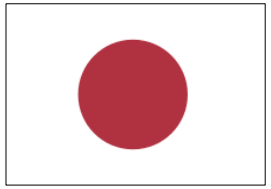
Segments

- Farms



AMBIENT SYSTEMS
Climate At Your Service

Aim to offer full-suite solution by Japan expansion

	2021	2022	2023	2024	2025
Market Expansion					
Offerings	Monitoring User Insights		Control		
Segments	Farms		Healthcare		Restaurants

AMBIENT SYSTEMS
Climate At Your Service

Received validation from farms

[CONFIDENTIAL]

Memorandum of Understanding

Comprehensive Climate Management System for Indoor Farming

Ambient Systems
Innovation 4.0
3 Research Link, #05-01
Singapore, 117602

Kingdom Farms
Singapore

November 5, 2020

Dear Mr. Timothy Tan,

Thank you for your interest in Ambient's climate management solution for indoor farms. At Ambient, we understand that managing the indoor environment can be a complex and time-consuming process. We want to make this easier for you by providing a comprehensive solution that handles all of your building management activities, allowing you to focus on your business.

As a deep-tech spin-off from National University of Singapore, Ambient's goal is to build solutions that will support urban farm operators with meeting their climate needs by leveraging technological advances in Internet-of-Things (IoT) and Machine Learning (ML). We propose to work together with Kingdom Farms on developing a climate management solution for your fleet of indoor farms. The motivation for this joint project is two-fold. First is to provide the necessary technology to effectively manage the indoor climate. Second is to leverage the collected data to develop a more precise method of indoor farming to achieve greater yield, higher quality, and more accurate harvest time predictions.

Scope of Work

Ambient's international team of engineers, software developers, and management consultants will help you obtain the most appropriate climate management solution for your needs.

The following services are offered as part of the overall solution:

Ambient Systems
i4 Building
3 Research Link #05-01
Singapore
117602

9th May 2021

Letter of intent

SINGROW
Growing Your Daily Fresh

Dear Ivan Damnjanovic,

I am writing to express our intention to do a pilot of Ambient's climate management solution at Singrow's indoor farm located in 102 Henderson Rd, Singapore 159562

Singrow is a Singapore-based company that develops innovative agritechnology solutions for producing premium-quality fruits and vegetables. Leveraging its proprietary platform, Singrow develops the freshest, best quality produce, such as non-GMO strawberry varieties in red and white aesthetics. More importantly, Singrow focuses its efforts on developing technologies that would help overcome ongoing challenges, such as global supply barriers, to meet the demand of the world's most discerning consumers.

Ambient understands that managing the indoor environment can be a complex and time-consuming process. Its mission is to make this process easier by providing a comprehensive solution that handles all building management activities, allowing indoor farm operators to focus on their core expertise in agritechnology.

Ambient proposes to work with Singrow on piloting a climate management solution at Singrow's indoor farm. The motivation for this project is two-fold. First is to provide the necessary technology to effectively manage the indoor climate. Second is to leverage the collected data to develop a more precise method of indoor farming to achieve greater yield, higher quality, and more accurate harvest time predictions.

We have received

2 LOI

from Kingdom Farms and SinGrow

Approx. Value for a 500 sq ft farm

SGD7.6K

About Us

- **MComp, Computer Science**
  National University of Singapore
- **BASc, Civil Engineering, Sustainable Energy**
  University of Toronto
- **Serial Entrepreneur**
  5 years experience in cleantech, deep tech

, Finance
Singapore Management University

- **Strategy and Fundraising**
  3 years experience across
  e-commerce, gaming and payments

AMBIENT SYSTEMS
Climate At Your Service

1. Experienced Team with tech, strategy and fundraising background

2. Received 2 LOI with indoor farms

3. Pilot test conducted with SinGrow resulted in actionable recommendations

4. Our Indoor Farming launch market growing at a CAGR of 24%

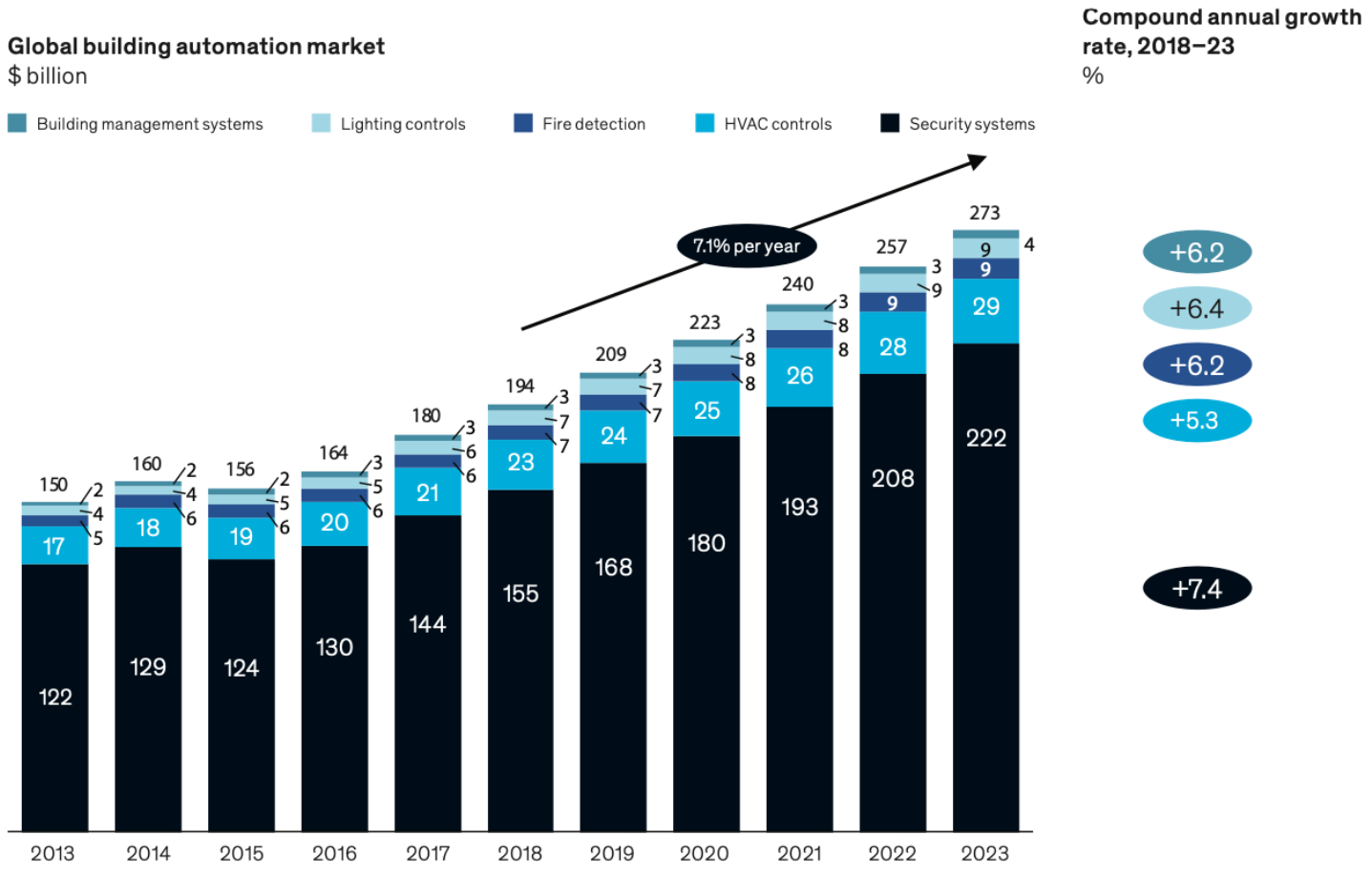
5. NPV of SGD8.5M with funding gap of SGD252K

Appendix

Total Addressable Market

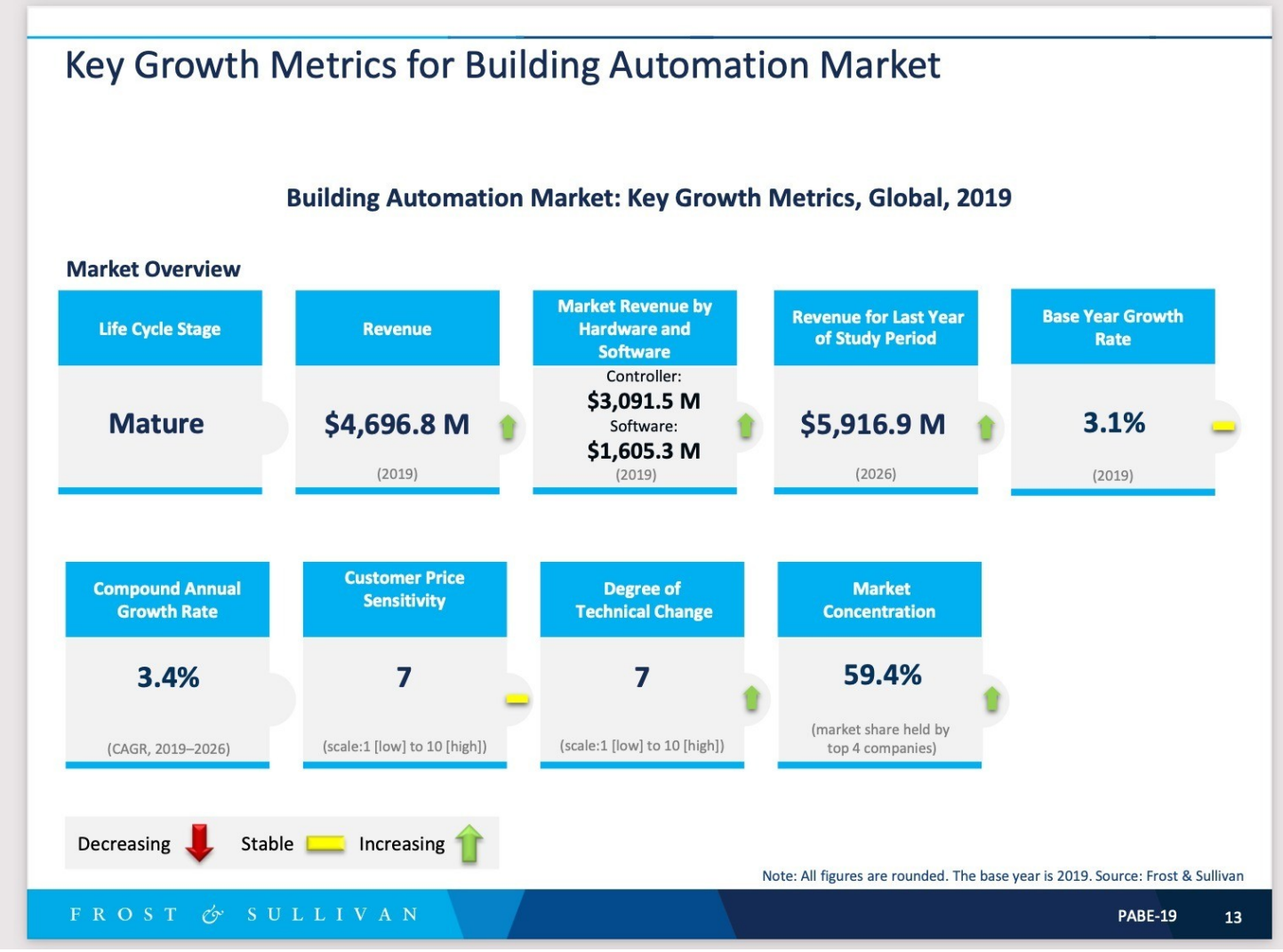
Exhibit 21

Building automation is projected to accelerate rapidly, particularly in security systems

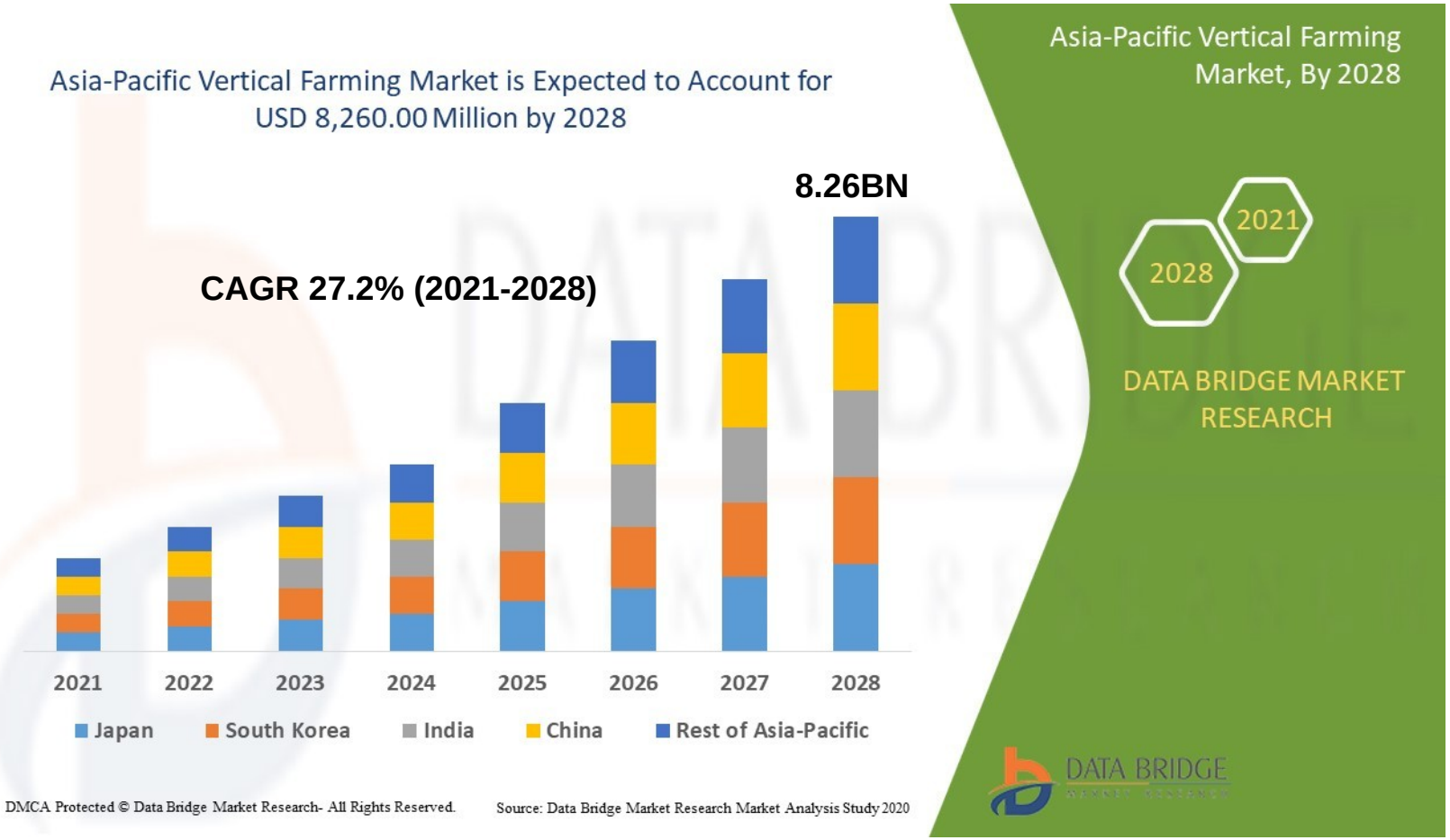


Source: Frost & Sullivan; IHS; Navigant

Serviceable Addressable Market



Serviceable Obtainable Market



DMCA Protected © Data Bridge Market Research- All Rights Reserved.

Source: Data Bridge Market Research Market Analysis Study 2020

Value-based Pricing

Source: Agfunder	Commercial Green House	Vertical Farming
Farm Size (SQFT)	280,000	67,000
Output (lbs)	2,000,000	2,000,000
Upfront Costs (USD/lbs)	0.61	1.3
Growing Costs (USD/lbs)		
Employee	1.1	1.57
Operating	0.42	0.42
Delivery Cost (USD/lbs)	0.2	0.2
Total Cost (USD/lbs)	2.33	3.07
Cost Differential (USD)		0.74
Cost Differential (%)		**24.10%**

1. We want to ensure price parity between operating a vertical farm and traditional greenhouse farms, in order to do that we need to bring price down by **24%**

	Min	Max
Cost of Employee (SGD)	$3,000	$4,000
Target Value Increase/pax	$723	$964
Monthly Value/sq ft	$1.45	$1.93

2. Tagging the **24%** increase in value to human capital productivity would mean monthly value derived/sqft to be around $1.45-$1.93

SinGrow		
Size (Sq Ft)	500	
Output (kg/harvest)	200	
No. of harvest/year	9	
Output (kg/year)	1800	
Weight of Punnet (kg)	0.2	
No. of Punnets/year	9000	
	Min	**Max**
Price/punnet	$9	$12
Annual Revenue	$81,000	$108,000
Monthly Revenue/Sqft	$13.50	$18

3. Monthly revenue/sqft from SinGrow's sales of strawberries is $13.50-$18/sqft

	Expected Value (%)	Min	Max
Energy Savings			
Decrease in utility costs	25%	$0.02	$0.02
Improved Crop Yield			
Increase in output	25%	$3.38	$5
Operational Efficiency			
Savings in man hours	25%	$0.39	$0.39
		$3.79	**$4.91**

4. Expected Value derived to be around $3.79-$4.91

Financial Model

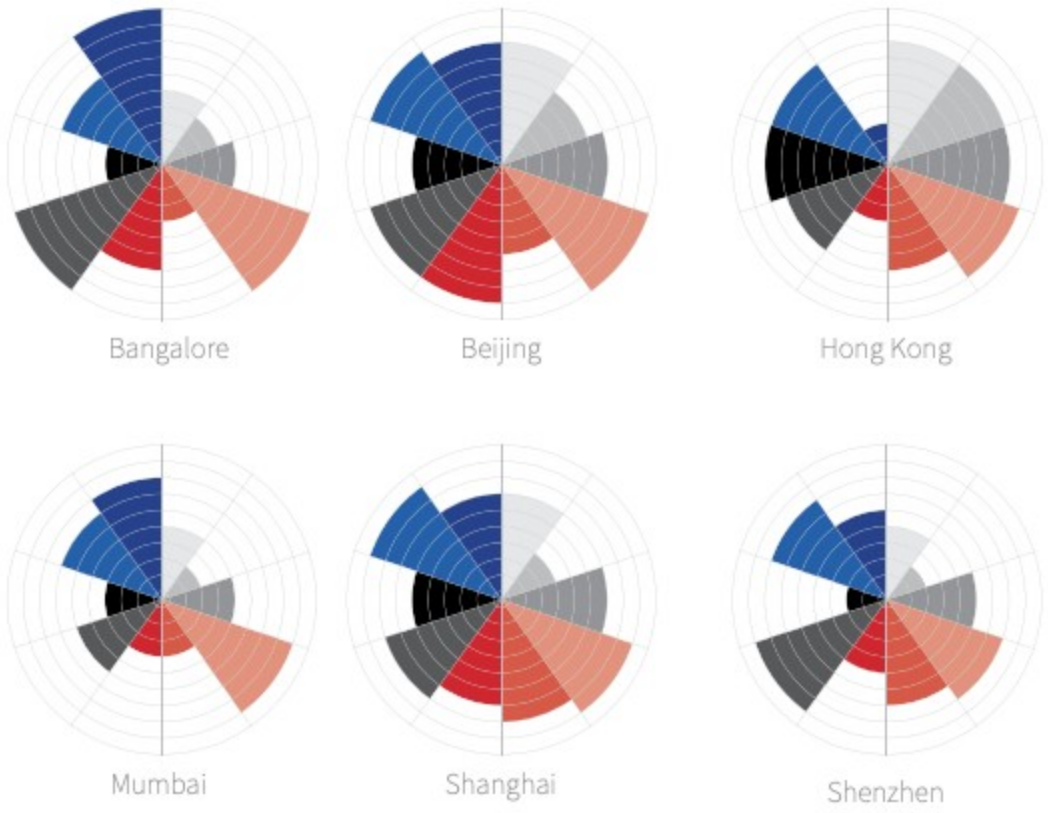
	Y1	Y2	Y3	Y4	Y5
Number of Customers managed	14	45	141	354	566
Area of Facility Size Managed (Sq Ft)	7,133	22,386	70,256	176,917	283,249
Ambient					
Revenue	$54,251	$488,319	$1,553,298	$3,961,237	$6,738,742
COGS	($14,597)	($107,885)	($262,720)	($448,786)	($561,349)
Gross Profit	$39,654	$380,435	$1,290,577	$3,512,451	$6,177,393
Profit Margin	73.1%	77.9%	83.1%	88.7%	91.7%
Operating Expense	($195,120)	($425,732)	($540,026)	($695,930)	($891,664)
P&L	($155,466)	($45,297)	$750,552	$2,816,521	$5,285,729
	($155,466)	($200,764)	$549,788	$3,366,309	$8,652,038
Discount Rate	8%				
Net Present Value	$8,483,135				

Use of GRIP Fund

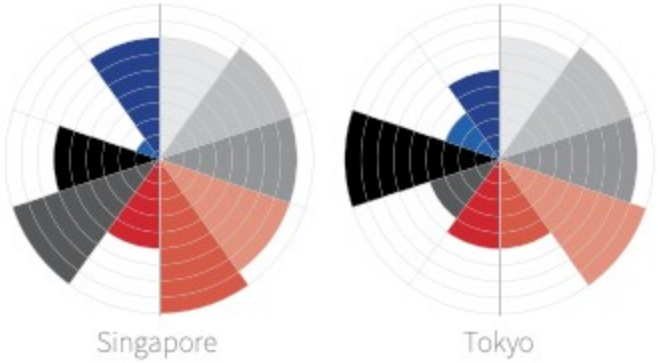
(S$)	Jul 2021	Aug 2021	Sep 2021	Oct 2021	Nov 2021	Dec 2021	Jan 2021	Feb 2021
Revenue								
Sensor Revenue	$8,571	$857	$943	$1,037	$1,141	$1,255	$1,380	$1,518
Integration Revenue	$0	$0	$0	$0	$0	$0	$0	$0
Software Revenue	$0	$0	$0	$0	$0	$0	$4,872	$5,359
	$0	$0	$0	$0	$0	$0	$0	$0
Data Insights Revenue	$0	$0	$0	$0	$0	$0	$0	$0
Total Revenue	**$8,571**	**$857**	**$943**	**$1,037**	**$1,141**	**$1,255**	**$6,252**	**$6,877**
Cost of Sensors	$8,057	$806	$886	$975	$1,072	$1,180	$1,159	$1,275
Cost of Integration	$0	$0	$0	$0	$0	$0	$0	$0
Cost of Maintenance	$0	$0	$0	$0	$0	$0	$0	$0
Total Cost of Goods Sold	**$8,057**	**$806**	**$886**	**$975**	**$1,072**	**$1,180**	**$1,159**	**$1,275**
Gross Profit	**$514**	**$51**	**$57**	**$62**	**$68**	**$75**	**$5,093**	**$5,602**
Operating Expense								
Rent	$0	$0	$0	$0	$0	$0	$0	$0
Tech Development	$4,500	$4,500	$4,500	$4,500	$4,500	$4,500	$6,000	$6,000
Salary	$6,000	$6,000	$6,000	$6,000	$6,000	$6,000	$12,000	$12,000
Others	$1,000	$1,000	$1,000	$1,000	$1,000	$1,000	$1,000	$1,000
Total Operating Expense	**($11,500)**	**($11,500)**	**($11,500)**	**($11,500)**	**($11,500)**	**($11,500)**	**($19,000)**	**($19,000)**
Profit & Loss	**($10,985.75)**	**($11,448.58)**	**($11,443.43)**	**($11,437.78)**	**($11,431.55)**	**($11,424.71)**	**($13,907.35)**	**($13,398.09)**
	($10,985.75)	($22,434.33)	($33,877.76)	($45,315.53)	($56,747.09)	($68,171.80)	($82,079.15)	($95,477.24)

Go-to Market Strategy

Future Asian agri-food tech investment and innovation hubs — relative strengths



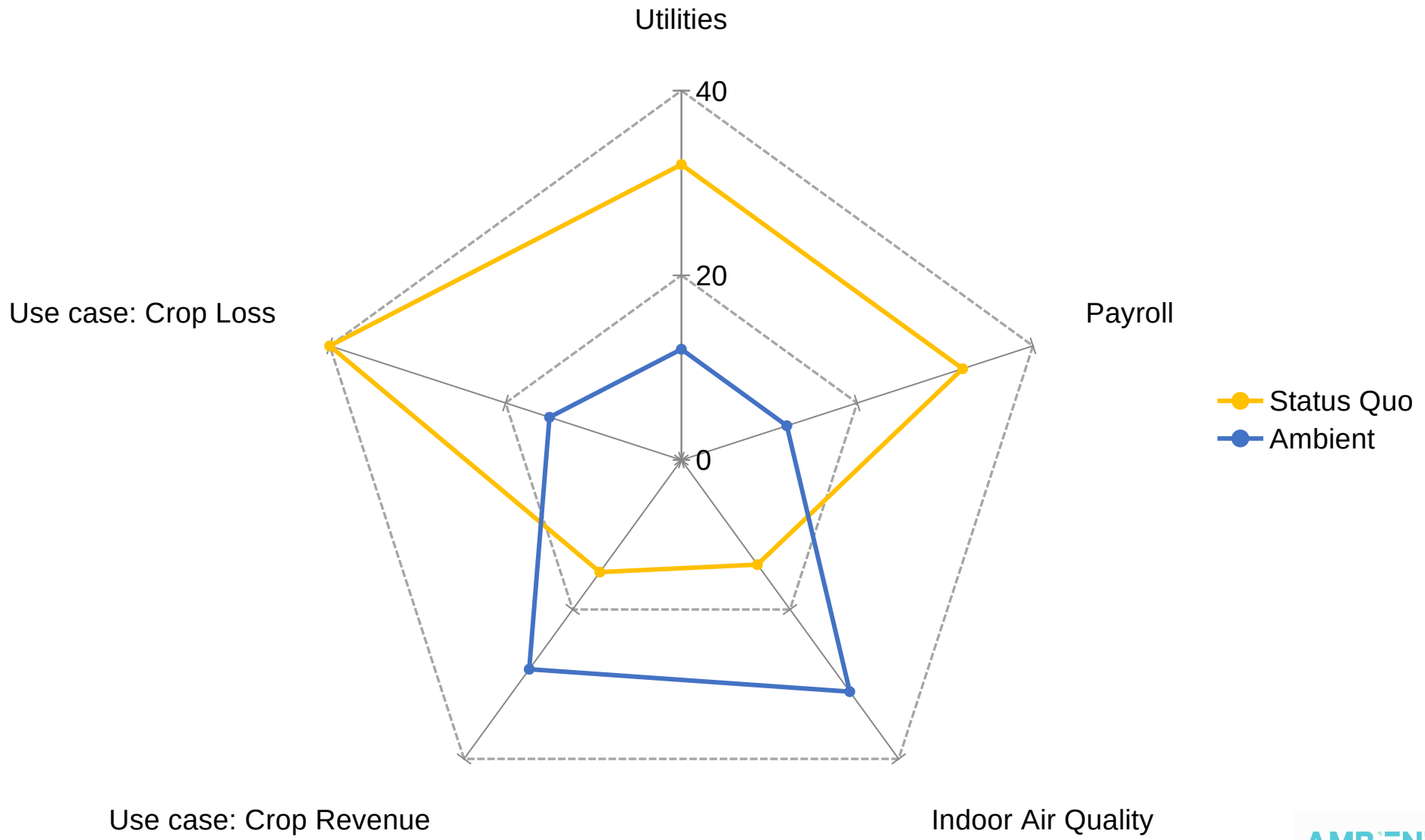
Number of Farms in Asia			
Country	Date	Type of Farms	Number
Singapore	2020	Leafy Vegetables	87
Hong Kong	2019	Organic Farms	322
Japan	2019	Indoor Farms	200
China	2020	PFAL	200



Sources: Capital IQ, Crunchbase, Economist Intelligence Unit, Eden Strategy Institute, INSEAD, Martin Prosperity Institute, Pitchbook, Quacquarelli Symonds, World Bank, PwC analysis

- Quality and engagement of academic institutions
- Talent competitiveness
- Quality of working and living environment
- Start-up ecosystem and entrepreneur culture
- Government initiatives supporting innovation and investment
- Existing presence of venture capital activities
- Technology availability
- Existing presence of large agri-food businesses
- Local market for agri-food technology
- Current levels of venture capital investment in agri-food tech businesses

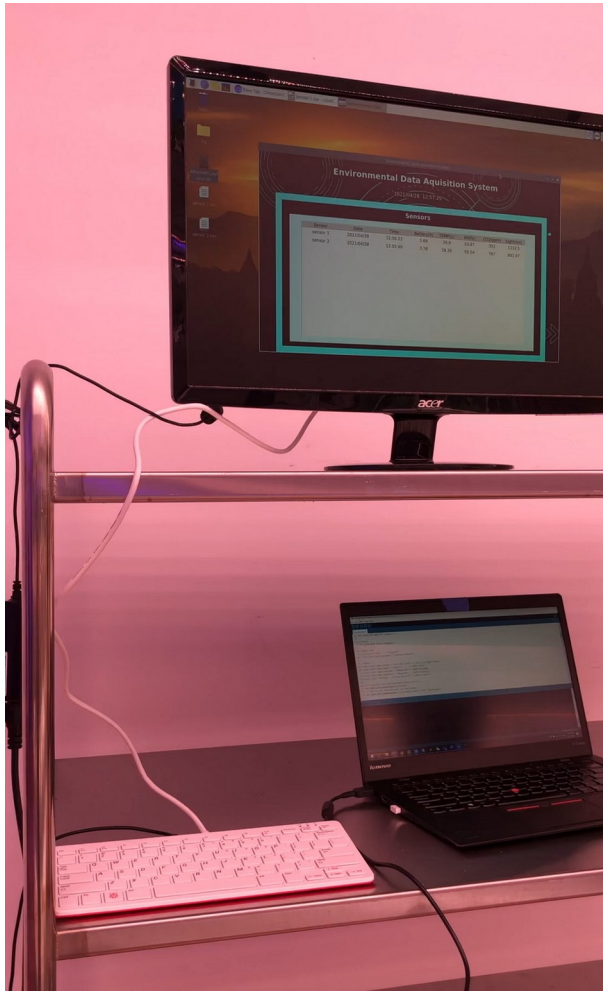
Problem: Multi-objective Optimization



Proof-of-Concept
MVP
Pilot
Outcomes

1. Collected climate data successfully
2. Identified gap between current and optimal
3. Applied s to determine optimal
4. Provided recommendations





Solution Targets Quantified

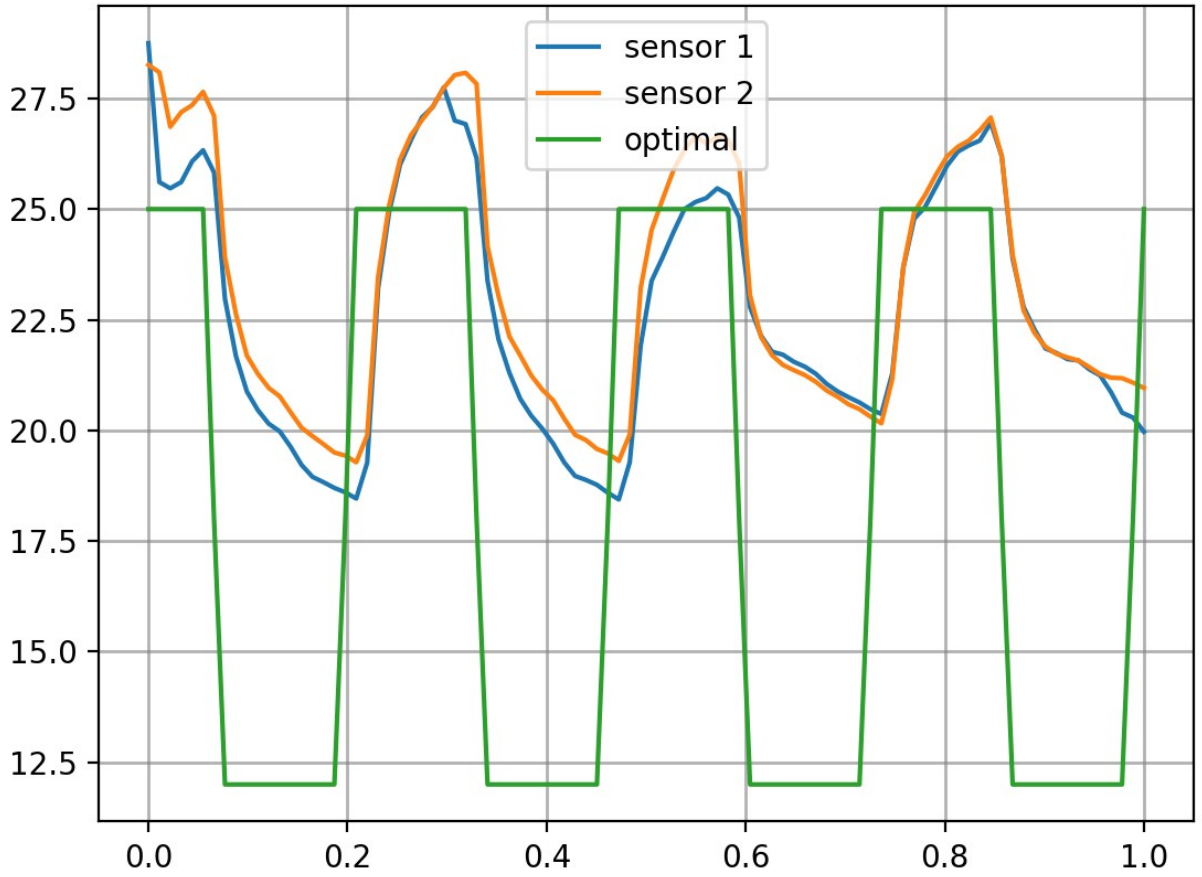
	Parameter	Actual	Optimal	Sample Action
Current POC	Air Temperature	28 C / 20 C	25 C / 12 C	Decrease temperature setpoint Increase ventilation
	Relative Humidity	55% to 70%	60% to 75%	Install fogging machines
	CO2	300 to 700 ppm	>330 ppm 500 to 1200 ppm	Decrease/increase ventilation (depends on CO2 concentrations outside) Install CO2 regulator
	Spectrum Light	2300 lux	5000+ lux	Adjust non-farming lights Increase light intensity Increase natural light

Solution Targets Quantified

	Parameter	Actual	Optimal	Sample Action
Current POC	Air Temperature	28 C / 20 C	25 C / 12 C	Decrease temperature setpoint Increase ventilation
	Relative Humidity	55% to 70%	60% to 75%	Install fogging machines
	CO2	300 to 700 ppm	>330 ppm 500 to 1200 ppm	Decrease/increase ventilation (depends on CO2 concentrations outside) Install CO2 regulator
	Spectrum Light	2300 lux	5000+ lux	Adjust non-farming lights Increase light intensity Increase natural light
Full-scale Plot	Soil Temperature	MVP 2.0		
	Soil Moisture			
	pH level			
	Electrical conductivity (EC_w)			
	Crop specific parameters			

Temperature

Leaf and petiole growth
Brighter leaf color
Greater sucrose
More biomass at the roots

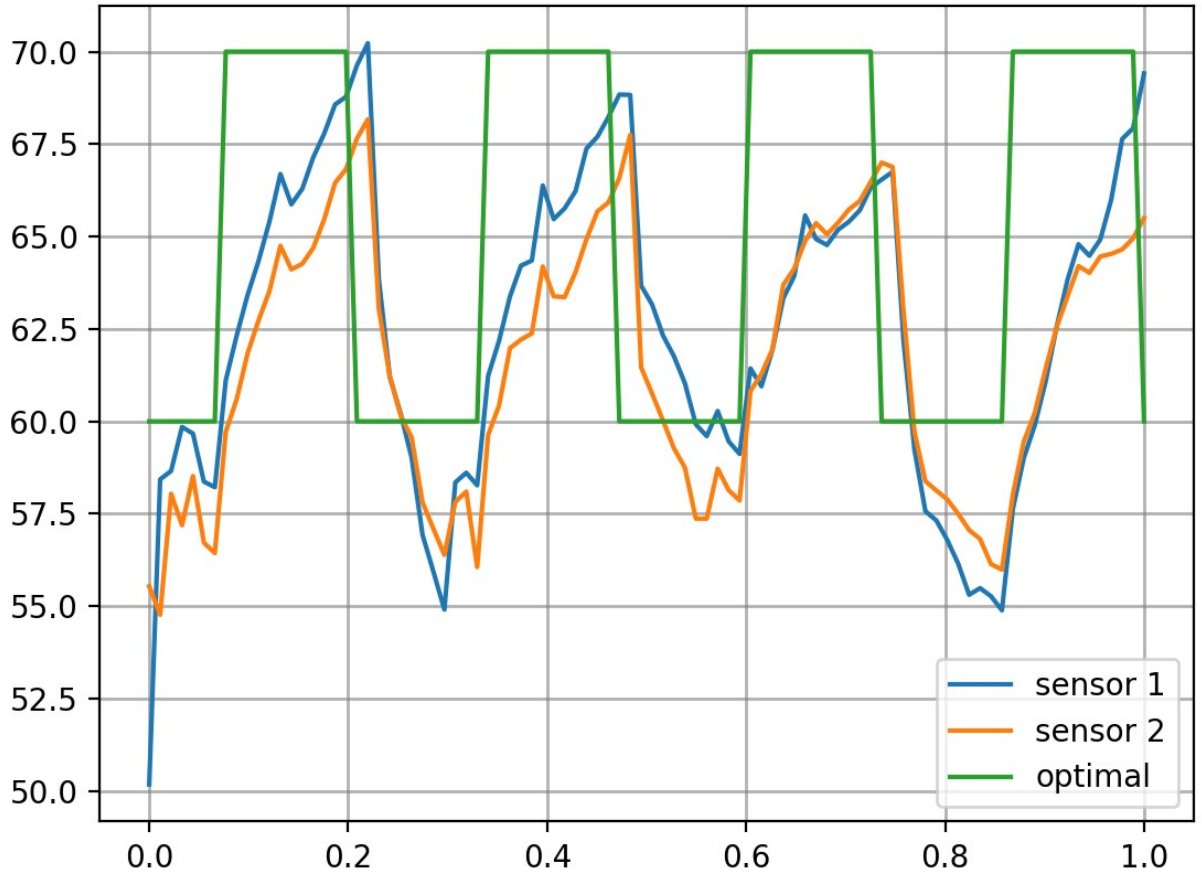


Temperatures after bloom affect plant growth and fruit quality of strawberry, , 2000

AMBIENT SYSTEMS
Climate At Your Service

Relative Humidity

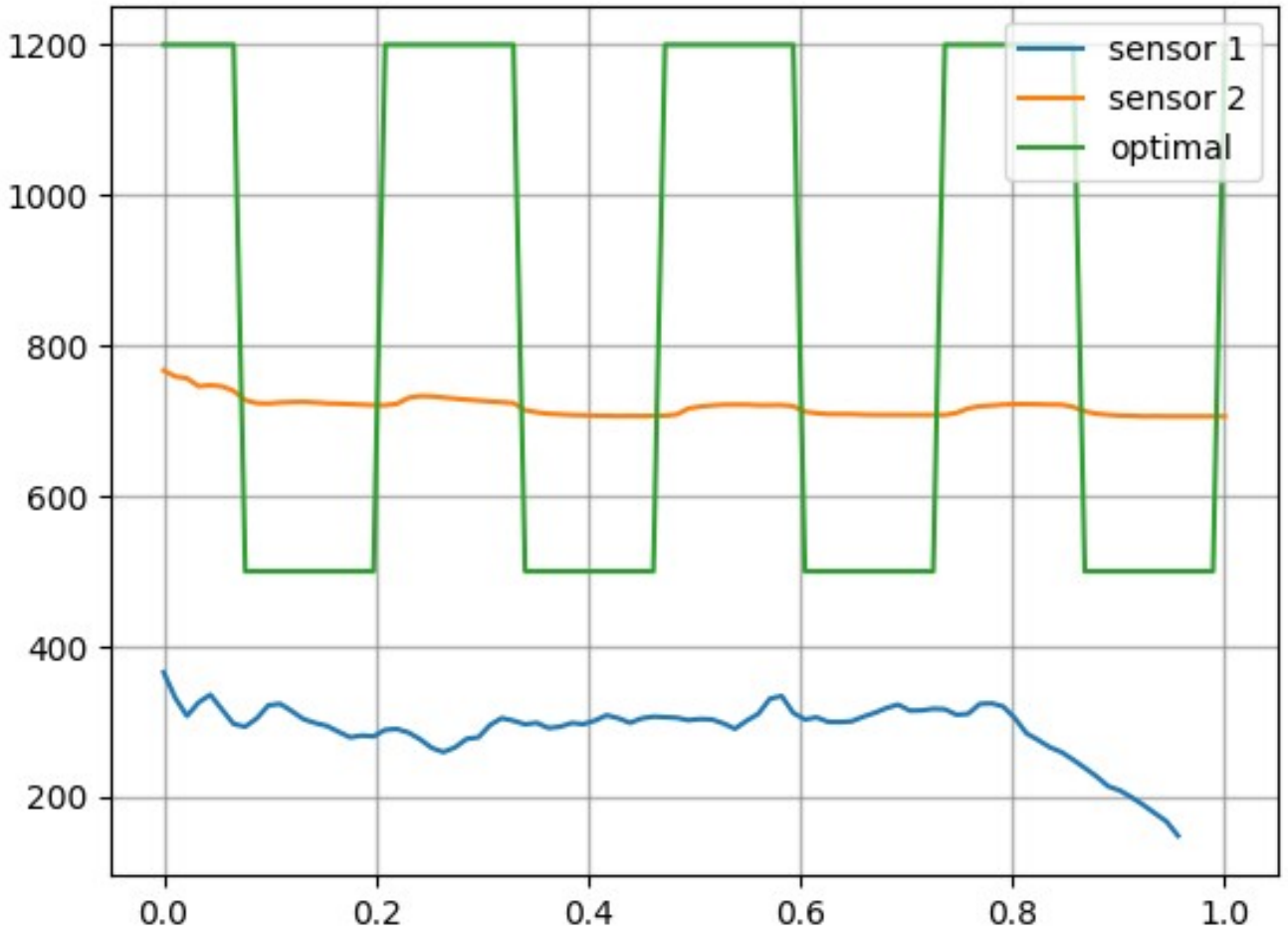
Avoid tip-burn and calyx burn
Avoid fungi
Avoid powdery mildew
Avoid leaf spot (bacterial blight)



Crop Guide: Growing Strawberry, , 2004

AMBIENT SYSTEMS
Climate At Your Service

CO2



Increased photosynthesis
Earlier flowering
Higher fruit yields
Improved stem strength and flower size

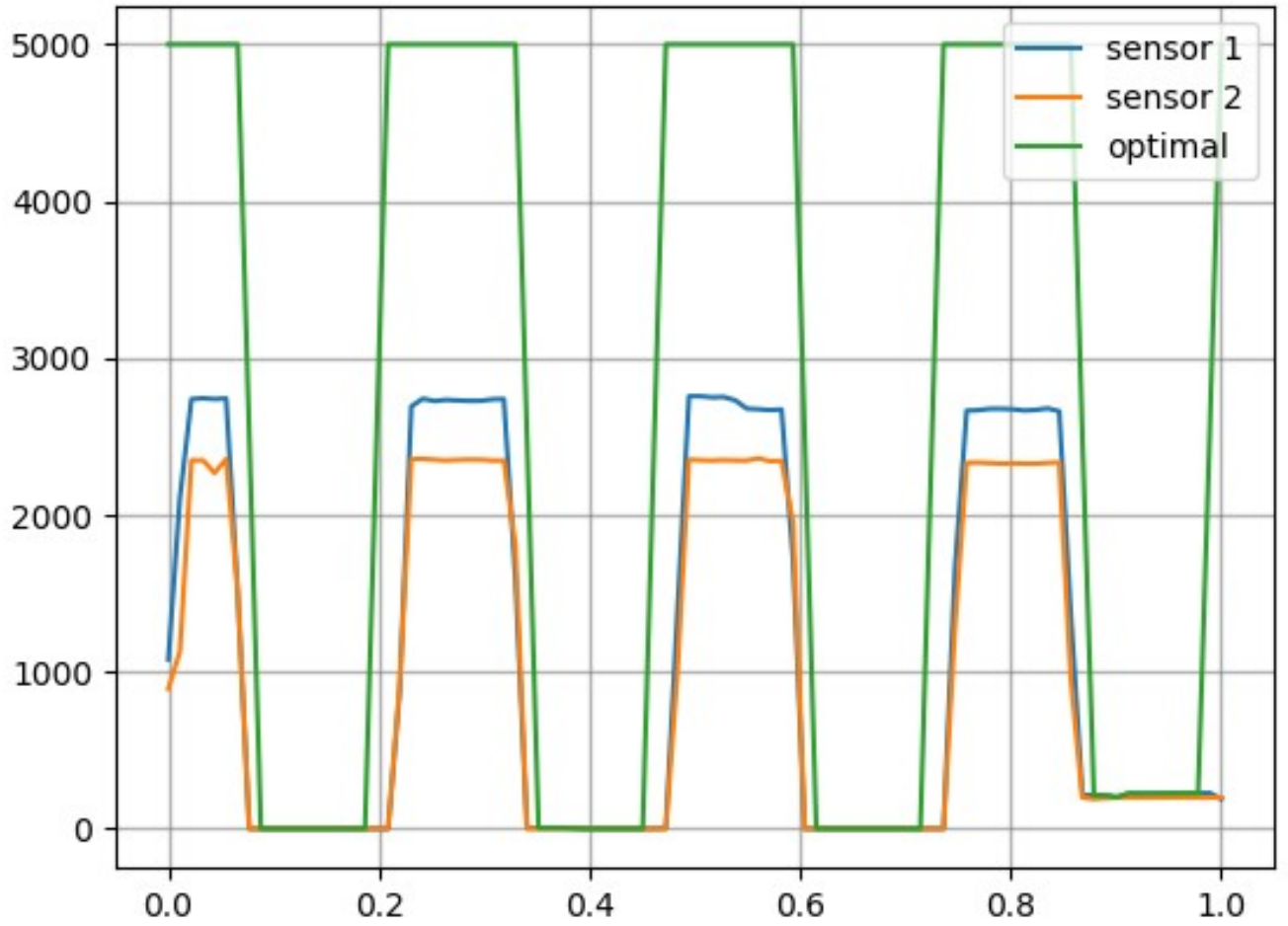
Carbon Dioxide In Greenhouses, Ministry of Agriculture, Food and Rural Affairs, 2021
Carbon dioxide (CO2) applications in indoor growing,

Private & Confidential

AMBIENT SYSTEMS
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Light Intensity

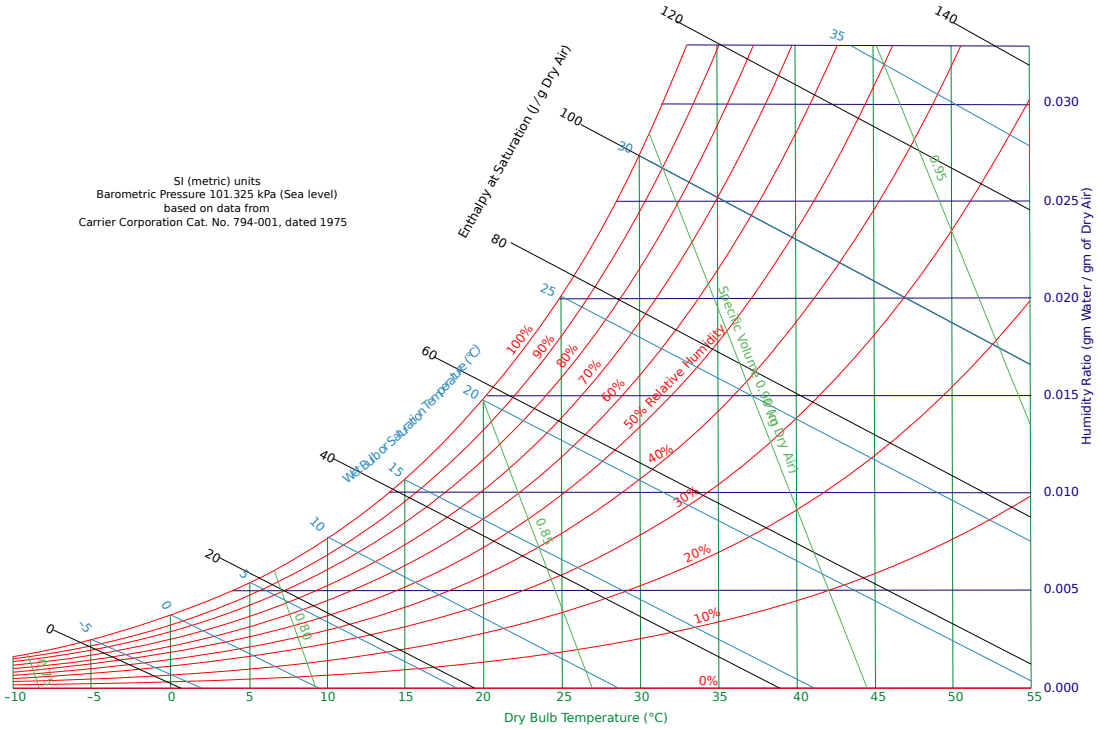
Increased photosynthesis



Effect Of Light Intensity, Temperature And Co2 Concentration On Photosynthesis And Growth Of In Vitro, International Society for Horticultural Science, 1989

#1: Interactions

- Psychometrics: temperature and relative humidity
- Light increases temperature and decreases humidity
- Decreasing temperature decreases CO2 concentrations in the air
- Increased ventilation introduces hot humid air from outside that needs to be cooled, purified, e hed, and re-humidified → very energy intensive → very expensive
- Safety: CO2 is needed by the plants, but human workers need oxygen to work indoors
- Photosynthesis process: lights and CO2 interact (both need to be high or both need to be low)
- Infiltration/exfiltration: porous building materials



#2: Equipment

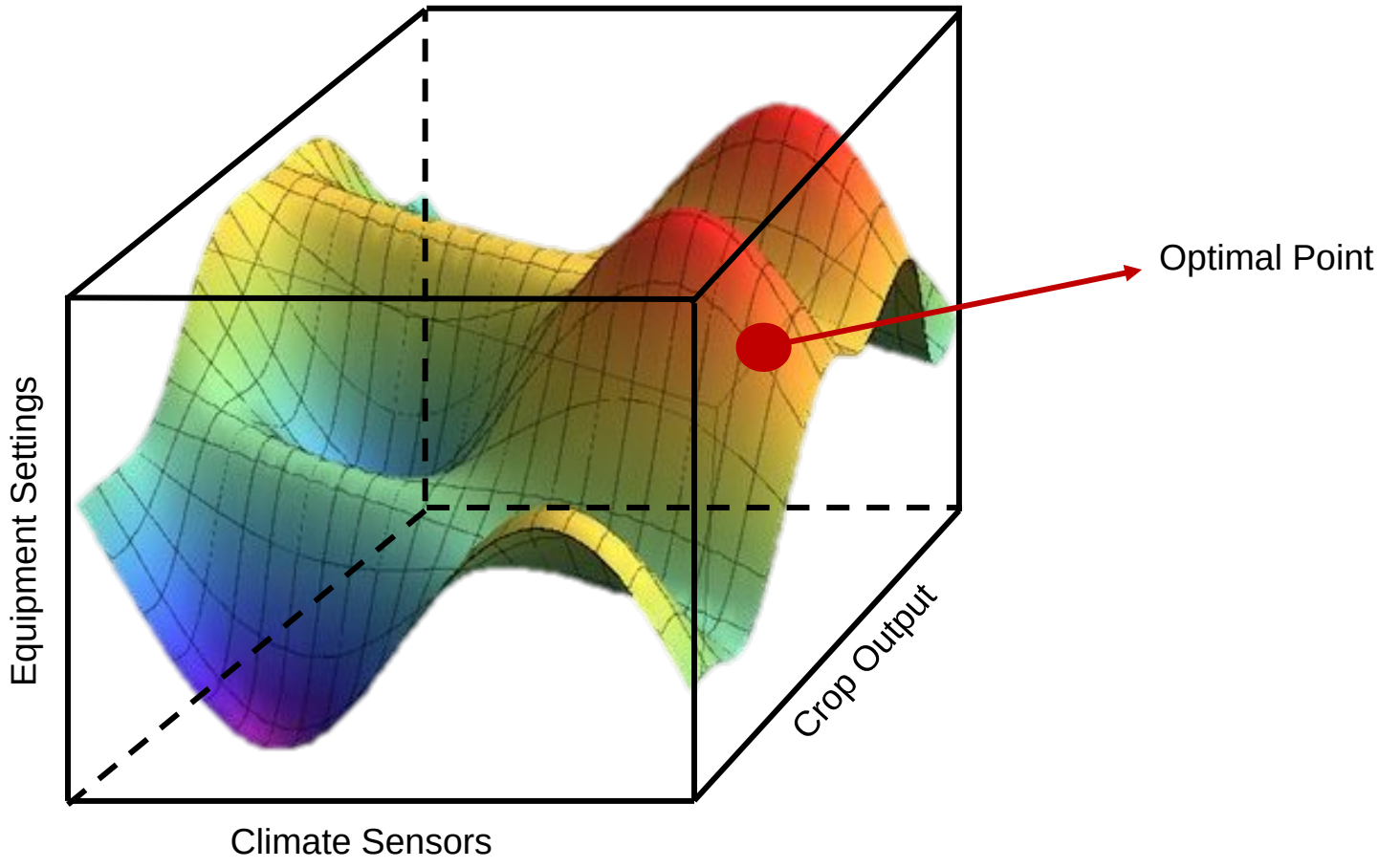
- Building equipment is often outdated
- AC controls are limited: only allows setting temperature and fan speed (not granular enough)
- AC controls cannot be managed remotely
- Need additional or custom equipment based on plant type, growing stage, and outdoor climate



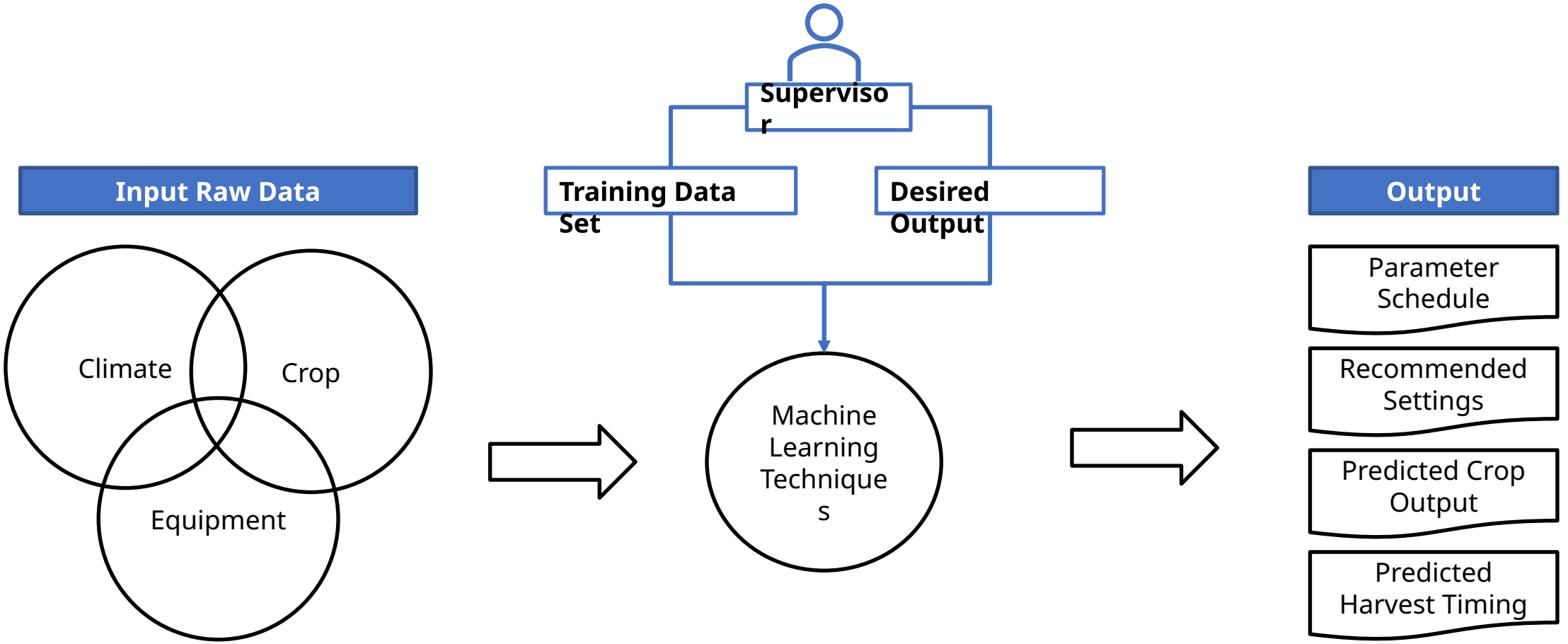




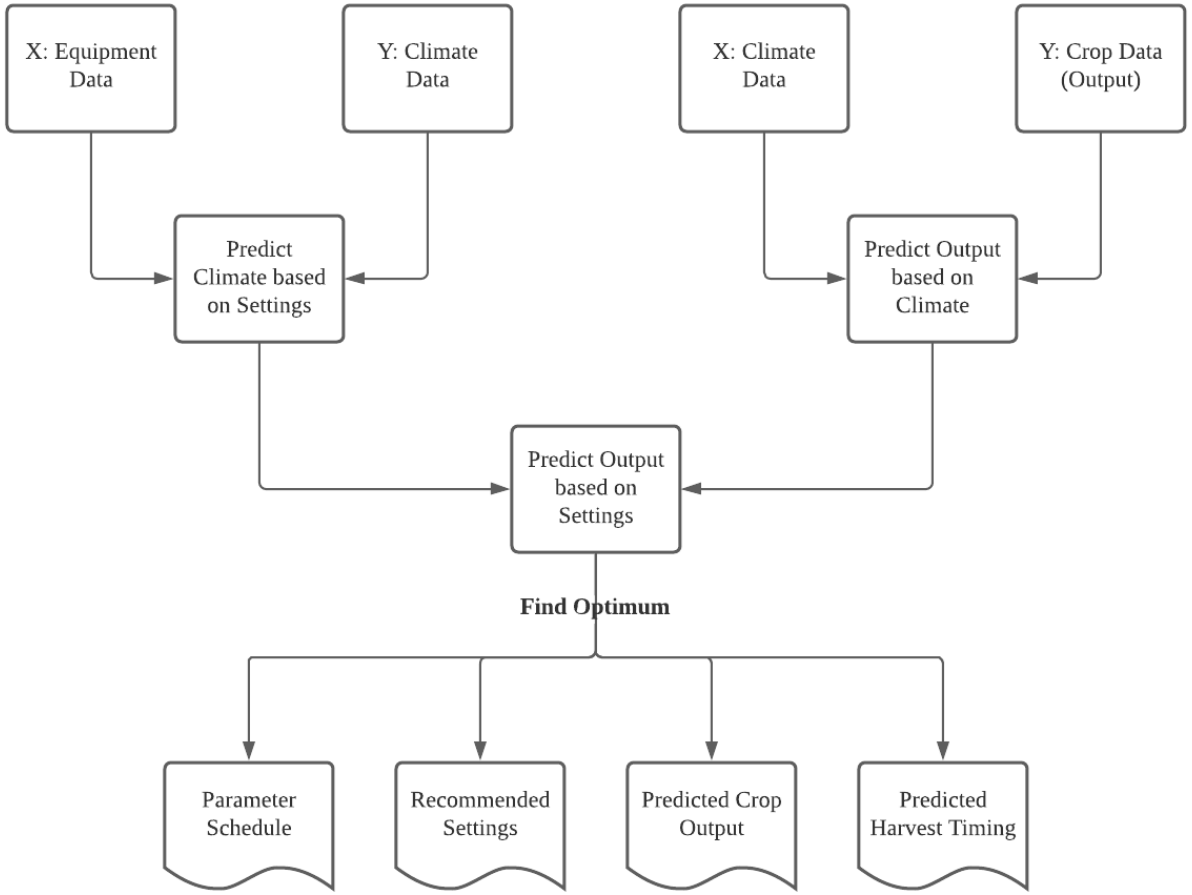
Optimization: Multi-Dimensional Gradient Descent



Ambient's Machine Learning Optimization Model



Supervised Learning: Three Stages



Notes:

- Non-climate parameters are already automated and controlled via dedicated systems (i.e., LED lighting, nutrition, irrigation, etc.)
- How Crop Data (output) is measured will vary based on farm type → it will impact data collected
- Product is agnostic to how crop output is defined or measured (assuming climate is an important input)
- More frequent data collection will increase the training of the feedback loop (e.g., providing crop data daily will allow quicker calibration than providing data at the end of each harvest)
- We can work with the farmer to provide the necessary data acquisition for crop quality (e.g., image recognition) and platform for training (e.g., optimization features)

Patent Classification

A	HUMAN NECESSITIES
B	PERFORMING OPERATIONS; TRANSPORTING
C	CHEMISTRY; METALLURGY
D	TEXTILES; PAPER
E	FIXED CONSTRUCTIONS
F	MECHANICAL ENGINEERING; LIGHTING; HEATING; WEAPONS; BLASTING
G	**PHYSICS** **G06:** COMPUTING; CALCULATING OR COUNTING **G06N:** COMPUTER SYSTEMS BASED ON SPECIFIC COMPUTATIONAL MODELS **G06N 3/00**: Computer systems based on biological models **G06N 5/00**: Computer systems using knowledge-based models **G06N 7/00**: Computer systems based on specific mathematical models **G06N 20/00**: Machine learning
H	ELECTRICITY

Productivity & Operational Efficiency

Ambient's optimization models show you how to create an optimal environment with:

- No need to manage sensors or raw data
- No need to manually adjust ACMV equipment early in the morning and late in the afternoon to accommodate plant growing schedule
- No need to manually adjust built-in (non-farming) lights
- No need for analysis of complex interactions between parameters
- No need to wait until seeing harvest results to fine-tune the parameters
- No need to deal with complex thermal inertia calculations
- No need for complex set-ups to perform farming experiments
- No need to spend time evaluating which mechanical equipment to buy (we recommend it)
- No need to worry about finding and selecting inter-operable systems (we shortlist them for you)

How much time can Ambient save you?

Recommendations

Insight	Recommendation	Cost
No historical data	Measure and securely archive critical operational data for crop repeatability and insurance	Ambient Platform Tier 1
Variability between days	Monitor for longer period of time (entire harvest cycle)	Ambient Platform Tier 1
No utilities data	Monitor energy consumption	$5k-$10k hardware modification Ambient Platform Tier 1 & 2
High temperature	Decrease daytime and night-time temperature	$5k - $10k portable AC
Low CO2	CO2 regulator (increase CO2)	$1k - $2k CO2 regulator $500 monthly CO2 supply
Moisture timing	Introduce mist earlier in the day Reduce mist later in the day	Ambient Platform Tier 1
Plants have various optimal parameter schedules	Determine farm portfolio optimal (not just for one plant but entire farm)	Ambient Platform Tier 1 & 3

Product Roadmap

Stream	V01 (Q1 2021)	V02 (Q2 2021)	V03 (Q3 2021)	Full Product Launch (Q4 2021)
Hardware	**Sensors** Collect essential parameters: Temperature, RH, CO2, and light intensity	**Sensors Extended** Collect additional parameters and increase number of sensors	**Collect crop data** Provide hardware to collect crop related data in real-time	Complete Sensors Controls
Dashboard	**On-Site Monitoring** Displays sensor readings in real-time	**Online Monitoring** Real-time visualization, analytics, and notifications	**Notifications** Enhanced visualization and notifications	Interactive and customized
Machine Learning	**Locally Deployed** Combination of real and simulated data	**Deploy on the cloud** Train using additional data and deploy throughh the online dashboard	**Fully Trained Model** Distribute training at the edge and in the cloud for real-time insights	Trained and deployed in real-time